10: Fingerprints Sampling

Stat 120 | Spring 2026

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```r
#| include: false

# Note: function written with gemini

library(magick)

#' Sample and Display Images
#'
#' @param image_paths A character vector of 3 image file paths or URLs
#' @param size The number of images to sample (e.g., 5 or 20)
#' @param probs A numeric vector of 3 probabilities
sample_fingerprints <- function(size = 5) {

  probs = c(0.08, 0.3, 0.62)

  image_paths <- c(
    "images/arch-green.png",
    "images/whorl-teal.png",
    "images/loop-purple.png"
  )


  # 1. Input Validation
  if(length(image_paths) != 3) {
    stop("Please provide exactly 3 image paths or URLs.")
  }
  if(length(probs) != 3 || sum(probs) != 1) {
    stop("Please provide exactly 3 probabilities that sum to 1.")
  }
  if(!size %in% c(5, 20)) {
    warning("This function was specifically designed for sizes 5 or 20, but will attempt to run
  }
```

```
  # 2. Read the images into R
  imgs <- c(
    image_read(image_paths[1]),
    image_read(image_paths[2]),
    image_read(image_paths[3])
  )

  # 3. Sample the indices based on the provided probabilities
  sampled_indices <- sample(
    x = 1:3,
    size = size,
    replace = TRUE,
    prob = probs
  )

  # Print the text result to the console so you can verify the frequencies
  # cat("Sampled sequence (Image Indices):\n")
  # print(sampled_indices)

  # 4. Extract the sampled images
  sampled_imgs <- imgs[sampled_indices]

  # 5. Format and display the images
  # Scale them to be uniform squares so the grid looks neat
  sampled_imgs <- image_scale(sampled_imgs, "200x200!")

  # Determine the grid layout: 5 columns x 1 row (for 5) or 5 columns x 4 rows (for 20)
  grid_layout <- ifelse(size == 5, "5x1", "5x4")

  # Create a montage (grid) of the sampled images with a 2-pixel border spacing
  img_grid <- image_montage(sampled_imgs, tile = grid_layout, geometry = "200x200+2+2")

  # Display the final graphic in the RStudio Viewer
  plot(as.raster(img_grid))
}
```

```
sample_fingerprints(size = 5)
```

> When you're done gathering your samples, input your sample statistics ($\hat{p}$) in this google sheet. Make sure to record your answer as a decimal point (e.g. 0.8 instead of 80% or 8/10).